STAT 641 **Midterm Exam 2 Solution** **Ghosh**

1. (a) (i) For all $x \neq 0, 1$, $X_n(x) = 0$ for large n, and $P(\{0, 1\}) = 0$. This proves $X_n \to X$ almost surely (P) as $n \to \infty$.
   (ii) Since for fixed $T > 0$, $\int |X_n| I_{\{|X_n|>T\}} \, \mathrm{d}P = n.m\big([0, \frac{1}{n}] \cup [1 - \frac{1}{n}, 1]\big) = n.\frac{2}{n} = 2$, $\forall\, n > T$, $\{X_n\}$ is not UI.
   (iii) Here $E|X_n - X| = E(X_n) = n.\frac{2}{n} = 2$ for all n large, so X_n does not converge to X in L^1.

   (b) If $\{X_n^p\}$ is UI, and the measure is finite (which is the case in the example above), then $X_n \to X$ a.e (P) would imply $X_n \to X$ in L^p for $p \geq 1$ (HW problem: 2.36(b)).There are other correct answers too, for example, along the lines of DCT.

2. (a) $\mu_3(\emptyset) = 3\mu_1(\emptyset) + 2\mu_2(\emptyset) = 0$; and $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A) \geq 0 \Rightarrow \mu_3(A) \in [0, \infty]$, $\forall\, A \in \mathcal{F}$; If $\{A_i\}$ is a disjoint collection of sets in $\mathcal{F}$, then $\mu_3(\cup_i A_i) = 3\mu_1(\cup_i A_i) + 2\mu_2(\cup_i A_i) = 3\sum_i \mu_1(A_i) + 2\sum_i \mu_2(A_i) = \sum_i (3\mu_1(A_i) + 2\mu_2(A_i)) = \sum_i \mu_3(A_i)$.
   So μ_3 is a measure.

   (b) $\nu(A) = 0 \Rightarrow \mu_1(A) = \mu_2(A) = 0 \Rightarrow \mu_3(A) = 3\mu_1(A) + 2\mu_2(A) = 0$. So $\mu_3 \ll \nu$.
   Will show: $\dfrac{d\mu_3}{d\nu} = 3\dfrac{d\mu_1}{d\nu} + 2\dfrac{d\mu_2}{d\nu}$ a.e (ν). Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\exists\, f_1, f_2 \geq 0$ s.t $\mu_1(A) = \int_A f_1 \, d\nu, \mu_2(A) = \int_A f_2 \, d\nu$. By definition, $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A) = 3\int_A f_1 \, d\nu + 2\int_A f_2 \, d\nu = \int_A (3f_1 + 2f_2) \, d\nu$ by linearity of integrals. Hence by uniqueness of RN derivatives, $\dfrac{d\mu_3}{d\nu} = 3f_1 + 2f_2 = 3\dfrac{d\mu_1}{d\nu} + 2\dfrac{d\mu_2}{d\nu}$ a.e (ν).

   (c) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $2\mu_2(B^c) = 0 = \nu(B)$, i.e. $2\mu_2 \perp \nu$. Choose $\mu_s = 2\mu_2, \mu_a = 3\mu_1$, then $\mu = \mu_a + \mu_s$. To show $\mu_a \ll \nu$ and compute $\dfrac{d\mu_a}{d\nu}$.
   Note that if $\phi_{2,3}$ represent the density of $Normal(2, 3)$, then by definition $\mu_1(A) = \int_A \phi_{2,3} dm$ and $\nu = m$. So, $\dfrac{d\mu_1}{d\nu} = \phi_{2,3}$ from RN theorem. Hence, from the proposition in the text, $\dfrac{d\mu_a}{d\nu} = \dfrac{d(3\mu_1)}{d\nu} = 3\dfrac{d\mu_1}{d\nu} = 3\phi_{2,3}$.

3. (a) Let $\phi(x) = -\log x, x > 1$, then $\phi(x)$ is a convex function.
   If $EX < \infty$, then since $0 \leq \log x \leq x, \forall x \geq 1$, we have $0 \leq E(\log X) \leq EX$, i.e. $E \log X < \infty$. By Jensen's inequality, $E\phi(X) \geq \phi(EX)$, i.e. $E(\log X) \leq \log(EX)$.
   If $EX = \infty$, then $E \log X \leq \infty = \log(EX)$ holds trivially.
   Hence $E(\log X) \leq \log(EX)$.

   (b) Apply Cauchy-Schwarz to $ZI_{\{Z>0\}}$ to obtain
   $$E[ZI_{\{Z>0\}}] \leq [E(Z^2)]^{1/2} [P(Z > 0)]^{1/2}$$
   using the fact that $I^2_{\{Z>0\}} = I_{\{Z>0\}}$ and $E[I_{\{Z>0\}}] = P(Z > 0)$. Also, since Z is nonnegative, $E(Z) = E(ZI_{\{Z=0\}}) + E(Z_{\{Z>0\}}) = E(ZI_{\{Z>0\}})$. Hence, we get
   $$[E(Z)]^2 \leq E(Z^2) P(Z > 0)$$
   which gives the inequality. Note $[E(Z)]^2 \neq 0$ here since $Var(Z) > 0$.